

Gate 5A - Review

Open only after saved attempt | RC1

How to use this review

Open this only after you have saved your attempt folder and evidence. Do not copy a finished answer. Use the review to diagnose the skill that failed, repair that skill, and then prove it again on fresh work.

Review dimensions

- Schema, grain and data-cleaning correctness
- Reference join and latest-record/window correctness
- Parquet/Delta partition and small-file reasoning
- Delta merge/upsert safety and rerun idempotency
- Shuffle/performance diagnosis and a justified mitigation
- Kafka topic, key, partition, consumer-group, offset and replay reasoning
- Structured Streaming checkpoint/state/output reasoning
- Technical explanation and defense of trade-offs

Repair workflow

1. Mark each failed checker/rubric item as DATA, SPARK, STORAGE, DELTA, PERFORMANCE, KAFKA, STREAMING or EXPLAIN.
2. Write one sentence describing the root cause. Example: "I partitioned by a near-unique business ID, creating too many tiny partitions."
3. Repair only the failed skill in a copy of your saved attempt. Do not overwrite the original evidence.
4. Rerun the checker and save the new output. A repaired attempt must still preserve correct earlier behavior.
5. Explain what changed, why it is safer/correct, and what would happen at 10x data volume.

Evidence to save

- Original saved attempt and checker output
- Repair notes with root cause and changed files
- Post-repair checker output
- Short defense covering grain, partitioning, merge/idempotency, offsets/replay, checkpoint recovery and one performance risk

Retest rule

After repair, attempt Gate 5B on the different fleet-sensor domain. Do not reuse A field names, outputs or code by mechanical substitution.

Passing a repaired A attempt does not replace the fresh retest. Gate mastery is recorded only after the required fresh-domain evidence also passes the critical floors.